

Mini Assessment

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Q1. If a database connection times out, what should your program do?

A) A connection timeout means the database server did not respond within the configured time. The program should handle the situation safely instead of crashing.

- Catch the appropriate timeout/database exception and prevent the application from stopping unexpectedly.
- Retry the connection only a limited number of times, using a short backoff delay between attempts.
- Set a sensible connection timeout so the program does not wait indefinitely.
- If all retries fail, record the error in the logs and inform the user with a clear message such as "Database service is temporarily unavailable."
- The program should fail gracefully and allow the user to try the operation again.

Q2. If cursor.fetchone() returns None, what should your program do?

A) None means that the query completed but no matching row is found. This should be treated as a normal "record not found" situation.

- Check whether the returned value is None before trying to access its fields.
- Return a clear "record not found" response or an appropriate default value.
- Do not index, unpack, or access attributes of None, because this can cause an exception.
- Log useful query or request context if the missing record needs investigation.
- Continue the program normally rather than treating the absence of a row as a database failure.

Q3. If a column value is missing from a query result row, why might .get() be safer than []?

A) When a database result is represented as a dictionary, using row['column'] can raise a KeyError when that column is absent. In contrast, row.get('column') can safely return None or a supplied default value.

- It avoids unnecessary KeyError exceptions for optional fields.
- A default value can be supplied, for example: row.get('email', 'Not available').
- It is useful when different queries return different sets of columns.
- The returned value can be checked before further processing.
- This makes database-handling code safer and more reliable when fields are optional or dynamically selected.